OS -- Operating System Study Notes Index

Format: Definition + Example + 40s script + Follow-ups. PSU MSTC interview ready.
OS definition was a mock interview gap -- read this file first.

Revision Order

Step 1   01-OS-BASICS-AND-PROCESSES.md   Definition, types, process states, PCB (10 min)
Step 2   02-THREADS-AND-SCHEDULING.md   Threads, ULT vs KLT, multithreading models (12 min)

Day-before quick revision: Cheat sheets at bottom of each file only.
OS technical definition is #1 priority -- say the 40s script aloud right now.

Files

File	Topics	Read Time	Status
01-OS-BASICS-AND-PROCESSES.md	OS definition, 8 types, OS services, 10 OS functions, Process in memory, 5 process states, PCB (7 fields)	10 min	[OK] Done
02-THREADS-AND-SCHEDULING.md	Thread definition, Thread vs Process (4 differences), ULT vs KLT, 3 Multithreading models, Advantages/Disadvantages	12 min	[OK] Done

Master Cheat Sheet

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OS Definition:
  "Software that acts as interface between user and hardware.
   Manages CPU, memory, devices. Takes user instructions   directs CPU   hardware acts."
OS Types (8): "Batch Multi Real Dist Embed Net Mobile Multi-proc"
  Batch   run jobs without user interaction
  Multi-tasking   multiple tasks on one CPU (time-sharing)
  Real-time   strict timing (RTOS)
  Distributed   multiple computers, appear as one
  Embedded   inside devices (car, microwave)
  Network   centralized resource sharing
  Mobile   smartphones (Android, iOS)
  Multi-processor   multiple CPUs
OS Functions (10):
  Memory, Process, File, Device, Networking, Security,
  Communication, Job Accounting, Secondary Storage, Command Interpretation
Process in Memory:
  Stack (top)   function params, return addresses, local vars (grows down)
  Heap     dynamic memory allocation (malloc/new)
  Data   global variables
  Text (bottom)   program code (read-only)
Process States (5): "New Ready Run Wait Terminated"
  New   being created
  Ready   waiting for CPU (in queue)
  Running   instructions executing on CPU
  Waiting   waiting for I/O or event
  Terminated   execution complete
Transitions:
  New   Ready: Admitted
  Ready   Running: Scheduler dispatch
  Running   Ready: Interrupt
  Running   Waiting: I/O or event wait
  Waiting   Ready: I/O or event completion
  Running   Terminated: Exit
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PCB (7 fields): "PP-PPC-RML"
  Pointer, Process State, Process Number(PID), Program Counter,
  Registers, Memory Limits, List of Open Files
Thread: lightweight process, shares parent's address space
  Has: Program Counter + Register Set + Stack Space
ULT vs KLT:
  ULT   user space, fast create, no kernel awareness, blocking = entire process blocks
  KLT   kernel space, OS manages, slower create, blocking = only that thread blocks
Multithreading Models:
  Many-to-One     many user   one kernel   no multiprocessing benefit
  One-to-One      each user   own kernel   true parallelism, expensive
  Many-to-Many    many user   &lt;=many kernel   BEST: flexible + no blocking
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Topics To Add Next

[ ] CPU Scheduling algorithms -- FCFS, SJF, Round Robin, Priority
[ ] Deadlock -- 4 conditions, Banker's Algorithm, Prevention
[ ] Memory Management -- Paging, Segmentation, Virtual Memory
[ ] Synchronization -- Mutex, Semaphore, Race Condition
[ ] File System -- Allocation methods (contiguous, linked, indexed)